

R PRADEEP KUMAR

DATA SCIENTIST

Results-driven Data Scientist, with over 3 years of experience in designing and implementing scalable ETL Pipelines, AI/ML Solutions and Big Data processing systems. Specialized in Data Analytics, Python, PySpark, ELK Stack, Monitoring & Observability. Expertise in leveraging Generative AI and LLMs to optimize workflows. Highly motivated about building innovative systems that drive decision making and operational efficiency.

Professional Experience

I. HCL Technologies, Chennai

Senior Analyst (R&D Engineer) - Research and Development

March 2024 – Present

- Designed and implemented ELK-based solutions for Log management and Observability, optimizing ELT pipelines for metrics analysis. Delivered multi-node ELK clusters deployed using Docker and Kubernetes orchestration, with cross-cluster replication and bi-directional cross-cluster search enabled. This solution significantly improved efficiency and ensured fault-tolerant.
- Engineered a fully automated PySpark ELT pipeline to transform RAS and IOC billing data into Delta Lake tables, exporting structured insights to Google Looker Studio for enhanced decision-making. Leveraged Python scripts for end-to-end automation.
- Leveraged Google Workspace for data storage in Google Spreadsheets, with preprocessing executed locally using PySpark. Implemented Python-based automation for efficient job scheduling. While Delta Lake ensured reliable and structured data storage. Created actionable insights through visualizations in Google Looker Studio, enhancing data-driven strategies and decision-making insights for HCL Software.
- Developed a Private, domain-specific Generative AI based LLM chatbot by integrating Large Language Models specifically using the Auto-Regressive Transformer models into Google Colab, enabling dynamic and efficient query handling capabilities by structured data queries using LangChain, RAG and Prompt Engineering techniques for efficient Q&A retrieval.
- Developed a Python automation script to monitor AWX job status and dynamically index job data into Elasticsearch for real-time observability. Implemented REST API integration, data filtering, and Elasticsearch indexing, enabling efficient log management and job tracking. Automated continuous polling with robust error handling, reducing manual monitoring efforts and enhancing system insights.
- Implemented Open Telemetry observability solutions integrated with ELK., Grafana, and ClickHouse, deployed using Kubemetes. Docker clusters, and manual setups on production servers.
- Experimented with diverse OTel Pipelines with Receivers, Processors & Exporters, utilizing manual and auto-instrumentation approaches, and deployed Open Telemetry Agent, Gateway, Collector, and Exporter implementations to enhance monitoring capabilities. Built a comprehensive OTel observability stack, reducing issue resolution times by 30%, improved system reliability and faster troubleshooting.

Personal Details

India Address

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Email Address

pradeep2496rk@gmail.com

Date of Birth

24th November 1996

Vellore

Technical Skills

Python

Apache Spark (PySpark)

Apache Hadoop

Databricks

Flask

Excel

SQL

NoSQL

Power BI

Looker

Elasticsearch

Logstash

Kibana

Grafana

Open Telemetry

Docker

Kubernetes

Shell Scripting

Apache Kafka

Git

AWS

LLM

Generative AI

- Trained interns and junior team members on Monitoring and Observability stacks, including AIOps frameworks, ELK multi-node clusters with APM, Grafana observability, and building custom Open Telemetry Exporters using Go modules for delivering advanced Open Telemetry observability solutions.
- Deployed Plane Project Management Application using Docker for agile issue tracking. Additionally, served as a Scrum Master, managing weekly sprints, assigning tasks, and overseeing the creation of deliverables for the team.
- Led cross-functional teams to implement AIOps framework using Python for automation and data observability workflows.

II. Inspirisys Solutions Limited, Chennai

Big Data Analyst (R&D Engineer) - Research and Development

March 2022 – March 2024

- Worked on Artificial Intelligence for IT Operations (AIOps) for Log based Anomaly detection for Network Streaming data for finding anomalies using Elasticsearch, Logstash & Kibana (ELK Stack).
- Integrated the LLM based AI Chatbots for generating responses which make use of Transformer architecture. These models have been implemented with LangChain framework using Retrieval Augmented Generation (RAG) for building a production level fully privatized dynamic chatbot which is capable of generating responses which fulfilled stakeholder's requirement.
- Integrated a streaming pipeline comprising of Apache Kafka, Machine learning models, PySpark and Grafana for dashboard monitoring real time streaming network event data.
- Worked on End-to-End Text generation based chatbots and also Voice chatbots integrations using large LLM's, engineered prompts based on different use cases and final model implementation in On-premise production servers.
- An ETL Pipeline using as Apache Kafka, Apache Spark, Delta Lake where streaming data from Twitter is taken as the Input and then pre-processed using Spark to transform the raw data into usable data for data ingestion using Delta Lake for Twitter Sentiment Analysis.
- Gained hands-on experience in using AWS services for deploying Machine Learning, Data Analytics projects and RESTful API's. Trained on Databricks projects using AWS.
- Expertise in integrating Hadoop and Spark Multi-Node Cluster for Data Analytics projects.

Achievements

- Awarded "Pat on the Back Certification" for best performance in Generative AI Projects on November 2023.

III. Itech Nemaraan, Chennai

Software Engineer Trainee - Internship

September 2021 – February 2022

Educational Qualifications

Post-Graduation – VIT University (Vellore Campus) (2021)

Master of Technology – (5 Year Integrated) Software Engineering.

CGPA - 7.77

Conference

AI Summit Databricks

AWS Innovate

Elastic TourOn Bengaluru